Experiment 4 : Class Diagram

Steps to create diagram

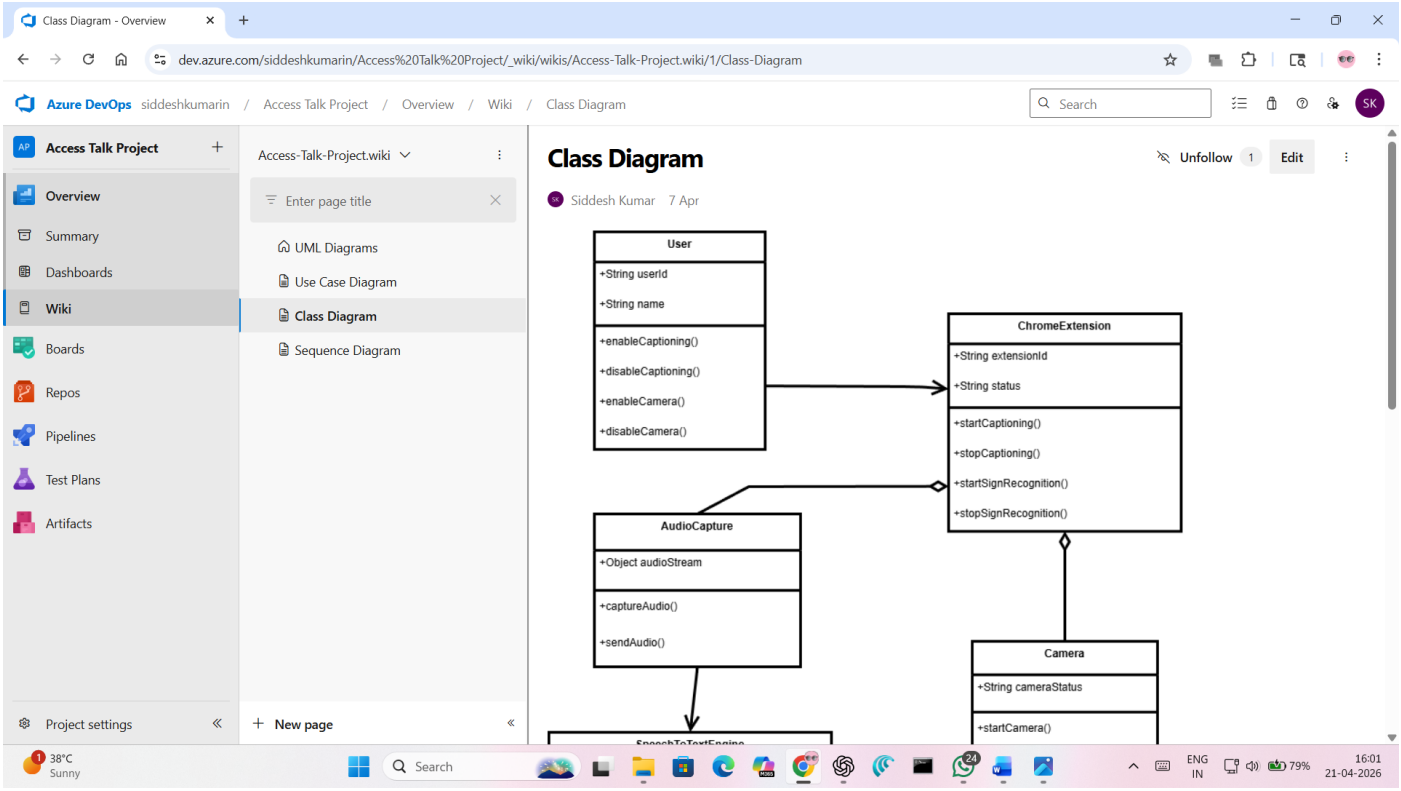
1. Open draw.io (diagrams.net)
2. Create Blank Diagram
3. Add class rectangles for:
   User, ChromeExtension, AudioCapture, Camera, SpeechToTextEngine, SignLanguageRecognitionEngine, TranslationService, TextDisplayUI, CaptionOverlayUI
4. Add attributes and methods inside each class
5. Connect arrows:
   a. Voice flow:
      User → ChromeExtension → AudioCapture → SpeechToTextEngine → TranslationService → CaptionOverlayUI
   b. Sign flow:
      User → ChromeExtension → Camera → SignLanguageRecognitionEngine → TextDisplayUI
6. Add relationships:
   User → ChromeExtension (association)
   ChromeExtension ◇ AudioCapture (aggregation)
   ChromeExtension ◇ Camera (aggregation)
7. Arrange layout:
   a. Left = Voice pipeline
   b. Right = Sign pipeline
8. Export as PNG

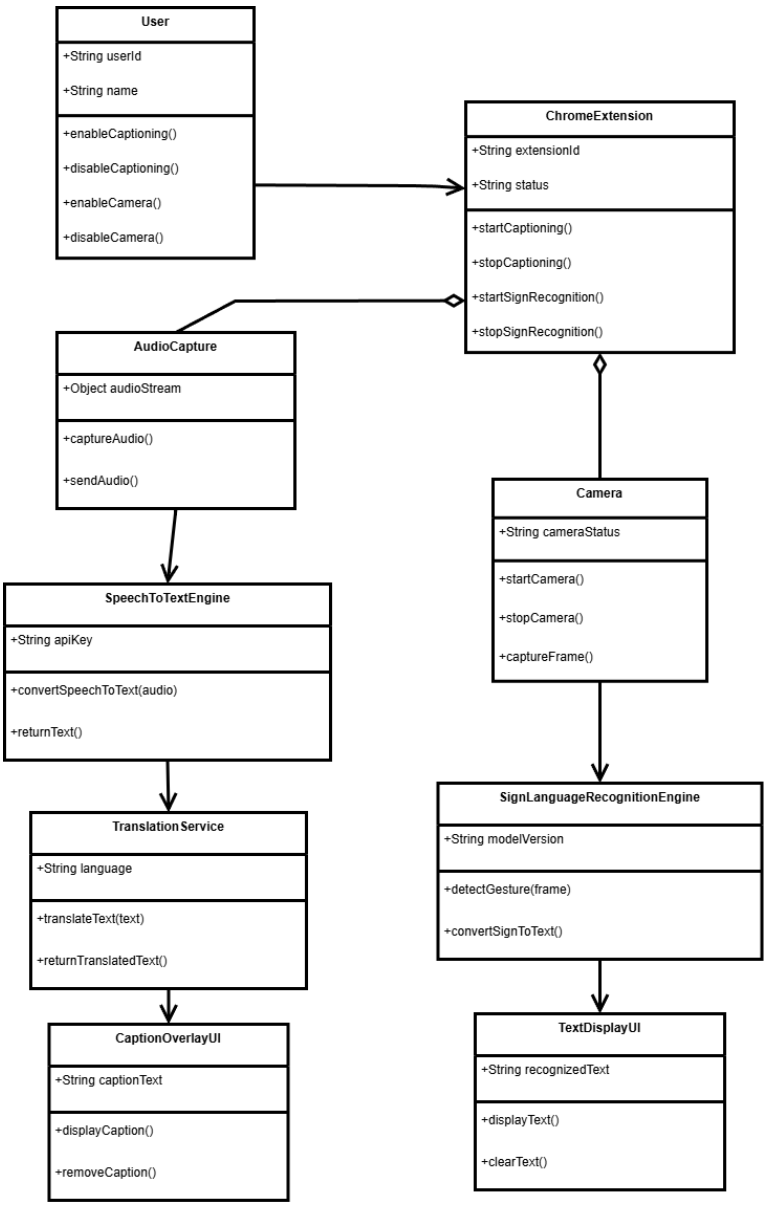
Diagram Notations

• User → controls system (enable/disable features)
• ChromeExtension → main controller of system
• AudioCapture → captures voice input
• Camera → captures video frames

• SpeechToTextEngine → converts speech to text
• SignLanguageRecognitionEngine → converts gestures to text

• TranslationService → processes translated text
• TextDisplayUI → displays recognized text
• CaptionOverlayUI → displays captions on screen


Class Diagram - Overview
dev.azure.com/siddeshkumarin/Access%20Talk%20Project/_wiki/wikis/Access-Talk-Project.wiki/1/Class-Diagram
Azure DevOps siddeshkumarin / Access Talk Project / Overview / Wiki / Class Diagram
Access Talk Project
Overview
Summary
Dashboards
Wiki
Boards
Repos
Pipelines
Test Plans
Artifacts
Project settings
Access-Talk-Project.wiki
Enter page title
UML Diagrams
Use Case Diagram
Class Diagram
Sequence Diagram
New page
Class Diagram
Unfollow
Edit
Siddesh Kumar 7 Apr
User
+String userId
+String name
+enableCaptioning()
+disableCaptioning()
+enableCamera()
+disableCamera()
ChromeExtension
+String extensionId
+String status
+startCaptioning()
+stopCaptioning()
+startSignRecognition()
+stopSignRecognition()
AudioCapture
+Object audioStream
+captureAudio()
+sendAudio()
Camera
+String cameraStatus
+startCamera()


User
+String userId
+String name
+enableCaptioning()
+disableCaptioning()
+enableCamera()
+disableCamera()
ChromeExtension
+String extensionId
+String status
+startCaptioning()
+stopCaptioning()
+startSignRecognition()
+stopSignRecognition()
AudioCapture
+Object audioStream
+captureAudio()
+sendAudio()
Camera
+String cameraStatus
+startCamera()
+stopCamera()
+captureFrame()
SpeechToTextEngine
+String apiKey
+convertSpeechToText(audio)
+returnText()
SignLanguageRecognitionEngine
+String modelVersion
+detectGesture(frame)
+convertSignToText()
Translation Service
+String language
+translateText(text)
+returnTranslatedText()
CaptionOverlayUI
+String captionText
+displayCaption()
+removeCaption()
TextDisplayUI
+String recognizedText
+displayText()
+clearText()